CONNOR GAG

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WORK EXPERIENCE

Machine Learning Scientist (Contract) — Feb 2026 - Present
HRL Laboratories — *Malibu, CA*

- Engineered end-to-end machine learning object detection and pose estimation pipelines for complex aerospace environments, leveraging transformer architectures to optimize inference speed and accuracy.
- Contributed to applied research involving multimodal systems and Vision-Language Models (VLMs), bridging the gap between spatial perception and language-based reasoning for autonomous applications.
- Translated theoretical AI research into deployable autonomous software by integrating state-of-the-art ML frameworks.

AI Researcher Intern — Jun 2025 - Sept 2025
NASA, Ames Research Center — *Mountain View, CA*

- Developed a Retrieval-Augmented Generation (RAG) Q&A chatbot that improved airspace incident analysis, reducing manual research time from several days to less than 60 seconds per query, thereby increasing operational efficiency.
- Authored an algorithm to automatically identify and structure potential JSON Traffic Management Initiatives (TMIs) from an entire day's worth of airspace planning videos, achieving 86% accuracy and enhancing data-driven decision-making.
- Optimized the Automated Speech Recognition pipeline, accelerating OpenAI Whisper transcription speed by 2x by converting to Faster-Whisper and improving the system's environment.
- Designed an architecture that supports real-time video analysis and transcription, facilitating faster decision-making for stakeholders and ensuring that 100% of video content was accessible within designated timeframes.
- After successfully pivoting the project to deliver a critical new feature, was selected by leadership to present the methodology and results to internal NASA teams on two separate occasions.

Graduate Student Researcher — Mar 2025 - Jun 2025
University of California San Diego, Cognitive Robotics Laboratory — *San Diego, CA*

- Engineered an offline, voice-controlled agent for the Spot robot by adapting NASA's ROSA framework to run locally on an RTX 2080 (8GB VRAM), eliminating cloud dependencies via 4-bit LLM quantization.
- Created an asynchronous perception pipeline using YOLOv8 and SmolVLM that decoupled heavy inference from the agent's reasoning loop, enabling real-time semantic understanding and 3D object localization.
- Architected a real-time sensor fusion pipeline in PyTorch, combining vision, depth, and skeletal tracking into a single score to enable autonomous user following with over 90% accuracy.
- Implemented a novel user acquisition module that creates a discriminative feature profile (appearance, depth, color) of a person in under 15 seconds, significantly outperforming general-purpose trackers.

Data Scientist — May 2022 - Jun 2024
UnitedHealth Group — *Remote*

- Developed an AI model to predict housing instability among Medicaid members, achieving over 92% recall, supporting healthcare interventions and improving member outcomes.
- Led data extraction and cleaning initiatives using complex SQL queries, processing medical claims for over 8 million Medicaid members, enhancing data quality and analysis accuracy.
- Designed a Retrieval-Augmented Generation system powered by OpenAI LLMs, enabling employees to query 1,000+ healthcare documents and 5 core tabular datasets using natural language.

EDUCATION

UC San Diego — Sept 2024 - Dec 2025
MS - Computer Science, AI Specialization — (GPA: 3.92/4.0)
Relevant Coursework: Natural Language Processing, Deep Learning, Probabilistic Reasoning, Algorithms

Gustavus Adolphus College — Sept 2018 - May 2022
BA - Computer Science, Minor in Math — (GPA: 3.96/4.0)

TECHNICAL SKILLS

AI Domains: NLP, RAG Architectures, Agents, LLMs (Llama 3, Qwen, GPT-4o), Quantization, Computer Vision
AI Frameworks & Libraries: Tensorflow, PyTorch, Hugging Face, LangChain, Whisper, Ollama, NLTK, OpenCV
Cloud & Platforms: Google Cloud (Vertex AI, BigQuery), Docker, ROS (Robot Operating System), MLOps, Snowflake
Programming & Data: Python, SQL, Bash, R, PySpark, Pandas, NumPy, Streamlit, Git, CI/CD, Linux